Aditee Jha

Chiragora, Hirapur, Dhanbad
+91-9122020011 aditeejha18@gmail.com
LinkedIn GitHub LeetCode CodeChef Portfolio Medium

Education

Birla Institute of Technology 2024 – 2028
B.Tech in Computer Science and Engineering — CGPA: 8.27 *Mesra, Ranchi*

Coursework

Data Structures and Algorithms DBMS Operating Systems Object-Oriented Programming Computer Networks

Projects

AgriGuard — GitHub — Live Demo 2026

- Built a full-stack smart agriculture platform using Next.js 16, React 19, TypeScript, Express 5, PostgreSQL, and Prisma 8, integrating satellite, weather, soil, and crop-health data into per-field irrigation and fertilizer recommendations.
- Engineered an FAO-56-based irrigation decision engine using crop water balance, rainfall forecasts, solar windows, soil moisture, and field parameters to determine when to water, wait, or skip; implemented Server-Sent Events for live updates and scheduled background analysis.
- Integrated Sentinel-2 satellite imagery and Open-Meteo weather/soil data for NDVI-based crop monitoring and field-level predictions; implemented a server-side MobileNetV2/ONNX disease classifier covering 38 crop/disease classes, Gemini-based explanations, Soil Health Card extraction, and AI second opinions.
- Designed an IoT-ready ESP32 field-node architecture with soil moisture, temperature, humidity, NPK, flow, energy, tank-level, and solar telemetry; implemented device authentication, pump control, sensor health checks, and automated irrigation event logging.
- Implemented a SHA-256 hash-chained sustainability ledger tracking water, energy, CO_2, and monetary impact, with measured/estimated status flags, automated validation, and role-based workflows for farmers, agronomists, and FPO administrators.

Spot2Solve — GitHub — Live Demo 2026

- Built a full-stack civic issue reporting and resolution platform using React 19, Node.js, Express, TypeScript, PostgreSQL, and Prisma, with separate Citizen, Authority, and Admin workflows and JWT-based RBAC.
- Engineered an end-to-end issue lifecycle with duplicate detection, geolocation, department auto-routing, SLA tracking, escalation, citizen verification, audit history, and role-specific dashboards for transparent issue resolution.
- Integrated Gemini for photo-based issue classification and severity detection, Cloudinary for evidence and voice-note storage, OpenStreetMap for reverse geocoding, and Leaflet for interactive maps, heatmaps, and proximity-based issue discovery.
- Implemented transactional status transitions with append-only audit history, background SLA monitoring with automatic escalation, notification fan-out via in-app alerts and email, and evidence-backed resolution workflows.

SafeDrop — GitHub — Live Demo 2026

- Built and deployed an AWS-powered e-waste management platform integrating device-specific data-wiping guides, safety-aware device assessment, recycler discovery, and collection-partner workflows.
- Designed a serverless architecture using AWS Amplify, API Gateway, Lambda, DynamoDB, Cognito, S3, Amazon Location Service, and Rekognition for hosting, APIs, data storage, authentication, location-based discovery, and photo-assisted functionality.
- Developed TrustChain, a traceability workflow using single-use handover verification, controlled lifecycle transitions, idempotency controls, and hash-linked event records to improve handover integrity and auditability.
- Implemented partner handover, weighing, dispatch, and facility-receipt workflows alongside an impact dashboard distinguishing recorded activity from estimated environmental benefits.

Technical Skills

- **Languages:** Java, JavaScript, TypeScript, SQL
- **Core Concepts:** Data Structures and Algorithms, Object-Oriented Programming, Database Design
- **Backend:** Node.js, Express.js, REST APIs, API Design, JWT Authentication, RBAC, Server-Sent Events
- **Frontend & Databases:** React, Next.js, HTML5, CSS3, PostgreSQL, MySQL, Prisma
- **AI/ML & Data:** Gemini, ONNX Runtime, MobileNetV2, Sentinel-2, Open-Meteo, FAO-56
- **Cloud & Tools:** AWS Amplify, API Gateway, Lambda, DynamoDB, Cognito, S3, CDK, Git, GitHub, Postman, Docker, Redis, Cloudinary, Leaflet, OpenStreetMap, Render

Coding Profile

Rated 2-star on CodeChef and 1750+ on LeetCode; solved 500+ DSA problems across platforms.

Achievements

- Qualified SIH 2026 College Round; worked on **MediKiosk**, a digital clinical intake kiosk that collects patient details, symptoms, and previous lab reports, generates concise clinical summaries for doctors, and uses OCR to extract important medical values.
- Winner of a college-level technical presentation on **AI Applications in Drones**.
- Winner of multiple college-level debate competitions.

Certifications

- IBM Full-Stack JavaScript Developer Professional Certificate – Coursera; Developing Front-End Apps with React – Coursera (IBM); The Complete Git & GitHub Course 2025 – Udemy.

Extracurricular

- Competitive debater and orator with top finishes across inter-school and inter-college events; known for analytical thinking and structured argumentation.
- Writer on Medium, translating complex ideas into clear, engaging narratives.